

Completed Pairs Hide Capped Failures: A ReVerPi Case Study of Selective Context Projection

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Abstract

Context projection replaces older tool observations with compact, addressable excerpts, reducing repeated input while potentially adding evidence-retrieval turns. We study this trade-off in ReVerPi, a Pi extension with archived observations and matched full/projected continuations. In an 86-run source-reading campaign with 641 model requests, the 15 completed pairs show identical success: 12/15 per arm. Twelve further boundary runs stop, with the runner suppressing the companion whenever the first arm fails to complete. Restoring all 27 boundary runs bounds projected-minus-full success between -9 and $+1$ tasks. One omitted, selector-chosen projected continuation successfully retrieves archive text yet exhausts twelve requests; its full counterpart answers in three. The eleven jointly correct pairs form a fully observed success stratum within this recorded frame: projection reduces aggregate logical tokens by 25%, while increasing the median pair’s tokens by 29% and total suffix requests from 35 to 55. Separating fitting from evaluation changes the selector’s apparent tie: outside its four fitting pairs, it incurs one extra failure and 8.6% more logical tokens over thirteen comparable runs. This methodological case study connects stopping rules, known bounded failures, unexecuted companions, and resource aggregation. Its findings concern the recorded campaign, rather than population noninferiority or superiority over unrestricted Pi. Evaluations should retain every intervention boundary, execute both allocated arms independently of the first arm’s completion, and report completion alongside interaction and token expenditure.

1 Introduction

A coding agent can send less information per request and still consume more work before producing an answer. Replacing an old observation may require the agent to search an archive, read an interval, and spend another model turn interpreting it. If that continuation reaches a resource limit, reporting only tasks that finish can remove the very cases in which projection is costly. The relevant question is therefore not merely how much input is removed, but how the changed interaction completes under a fixed allowance.

We examine this question in *ReVerPi*, an extension to Pi, a minimal terminal coding harness with public TypeScript extension interfaces and an RPC execution mode [1]. ReVerPi preserves the recorded history and replaces eligible tool observations only in outgoing requests. A content-addressed archive supports literal search and exact reading. Its controlled, read-only adapter seals a common prefix and compares full and projected continuations at the same first request boundary. These mechanisms build on established agent interfaces, observation management, and paired continuation [2–5].

The empirical basis is an adaptively developed source-reading campaign: 76 paired-or-capture submissions and ten single-arm integration runs. Fifteen runs complete both arms, and each arm is correct in twelve. Taken alone, this completed-pair view looks balanced. However, twelve additional runs reach the intervention boundary and stop. The controller aborts the pair when an arm fails to complete, so first-arm caps deterministically suppress companion execution. A frozen selector chooses projection in an omitted example, *pathspect-util*: full observations yield a correct answer after three suffix requests, whereas projection reaches its twelve-request cap despite successful retrieval of archived text.

Three distinctions change the interpretation. First, a cap without an answer is a known failure of completion by that cap; an unexecuted arm has an unknown result. Second, the eleven jointly successful pairs are completely identified within the 27 recorded boundary runs, because every other run has at least one known failure. Their cost contrast is meaningful for this stratum, even though it does not describe the whole frame. Third, the selector’s equal failure count on seventeen comparable runs includes its own fitting cases. On the thirteen runs outside fitting, it incurs an additional failure and higher expenditure. These distinctions concern the same evidence, not new tasks, new model calls, or a retuned method.

Position and contributions. This is a *methodological case study* of selective context projection. It contributes (i) a reconstruction of the stopping and observation process that explains why completed pairs omit relevant capped outcomes; (ii) a stratified analysis of fitting cases, first-arm outcomes, joint success, and request-level resource use; and (iii) an application

of established partial-identification and stratification tools to an executable agent protocol. The result is a concrete account of how a resource-saving intervention is measured. The scope of supported claims is consolidated in Table 6; implementation history and administrative provenance are kept in the appendix.

Code and data. Project code is hosted at https://github.com/timwhitez/ReVer_Pi. The source archive accompanying this article includes an `anc/` directory with analysis data, reproduction scripts, and the full displayed observation; Section E specifies its contents and the historical implementation identity.

2 Related Work

Agent interfaces and managed memory. SWE-agent studies how the agent–computer interface affects software-engineering behavior [2]; its history processors include observation-window management. OpenHands provides a general software-agent platform, with condenser interfaces that can replace older events with summaries [6, 7]. MemGPT manages information across memory tiers to extend the effective context available to an agent [8]. SoL-Pi combines capability-constrained harness research with observation archiving, paged recall, compaction, and other efficiency mechanisms [3]. ReVerPi belongs to this family: its archive and projection are the setting in which we study evaluation, rather than a newly invented memory primitive.

Prompt and trajectory compression. LLMingua compresses prompts with a budget-aware procedure, while Selective Context filters information using self-information [9, 10]. Lost in the Middle shows that the location of evidence affects long-context use [11]. In an interactive trajectory, however, a removed passage may later cause another tool call or model request. The Complexity Trap compares observation masking, summaries, and their combinations [4]; ACON optimizes compression guidance from feedback [12]; TRACE evaluates compression through closed-loop continuations from common states [5]. We use matched continuation with a narrower read-only replay contract and analyze what happens when one continuation is never executed.

Cost and quality evaluation. AI Agents That Matter argues for evaluating accuracy together with cost [13]. Token Reduction Is Not Cost Reduction studies cache and interaction effects on economic conclusions [14]. Robust aggregation is also central to empirical sequential-decision evaluation [15]. Our analysis distinguishes request counts, mean tokens per request, shared-prefix cost, and cached-input categories; sum, geometric-mean, and median ratios answer different questions even within a fixed successful stratum.

Missing outcomes and post-intervention strata. Manski’s worst-case bounds characterize partial information without

filling unobserved outcomes [16, 17]. Missing-data and positivity theory distinguish explaining an observation process from identifying outcomes in a region with zero observation probability [18, 19]. Principal stratification compares outcomes within strata defined jointly across treatment conditions, including outcomes truncated by death [20, 21]. Lee’s trimming bounds address selection under additional assignment and monotonicity conditions [22]. We make a limited, explicit connection: joint success is fully observed for the recorded paired continuations, while a population always-success effect under newly sampled stochastic continuations is a different estimand. Lee-style trimming assumptions are not imposed on this adaptive campaign. Standard risk-control and sequential-inference results provide the background for the appendix’s calibration discussion [23–25].

3 System and Evaluation Protocol

3.1 Outgoing projection and archived evidence

Table 1 summarizes the recorded configuration and resource limits. ReVerPi separates the intact history H_t from the request view $P(H_t)$. A result becomes eligible after appearing in two completed full-observation requests, provided it is at least 10 KiB, is outside the protected most-recent result, and is neither an error nor an archive-recovery or revalidation result. Eligibility refers to recorded transmission, not to the model’s comprehension. The configured excerpt budget is 1,024 bytes, divided between the original head and tail. A stable content hash identifies the complete archived observation.

The model can call `search_evidence` to locate a literal substring or `recover_evidence` to read an exact interval. Search returns original-text excerpts as well as handles, so it may provide sufficient evidence without a subsequent exact read. Both operations consume the same three-call allowance. Revalidation is a separate capability of the broader framework; the source-reading campaign uses only reading and archive access. Figure 1 shows the system and its historical evaluation controller. A genuine replacement and a task example appear in Section D.

3.2 Source-reading tasks and common-prefix replay

Questions are investigator-authored over fixed subsets of Python package sources (Table 12). The adapter exposes a bounded read tool over allowlisted files, without shell execution, writes, or network tools. Answers are controller-side JSON objects with exact keys and typed values; Section D supplies a complete question and answer-format example. The archive’s integrity and file allowlist support the experimental contract, not an adversarial operating-system sandbox.

Capture sends full observations until the first eligible request is prepared, then pauses before dispatch and seals the completed request/response tape. Full (F) and projected (P) continuations replay the same tape. Their first actual suffix

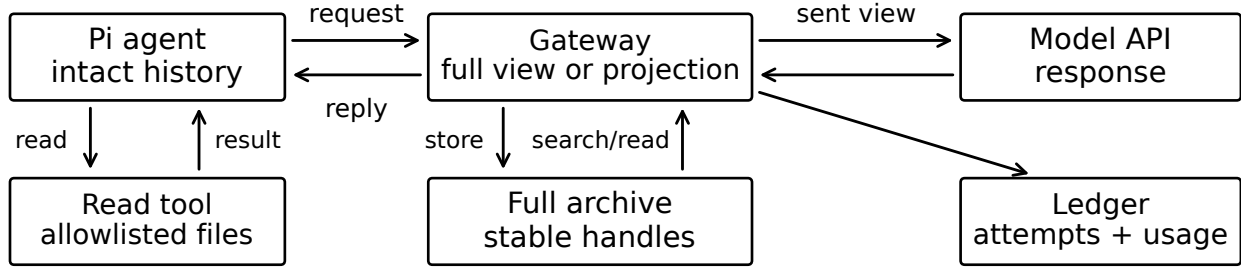
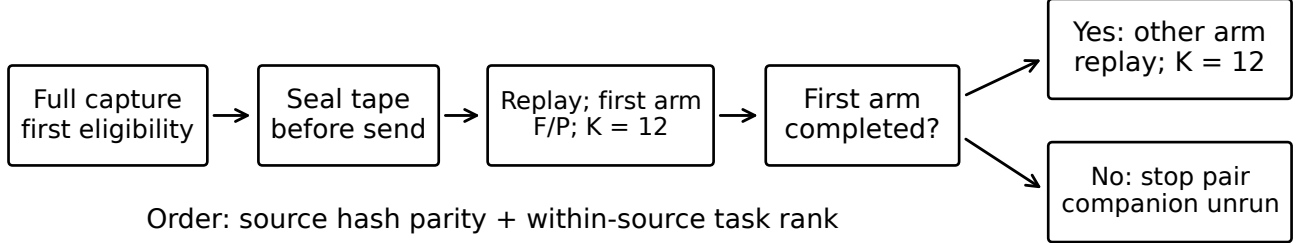
(a) Request projection and evidence access**(b) Historical two-arm scheduling**

Figure 1: ReVerPi and the evaluated controller. (a) The gateway changes eligible observations in outgoing requests while retaining exact archived content; the ledger records model attempts and usage. (b) Capture pauses before the first eligible dispatch, and each branch replays the completed tape. The historical controller runs the companion only if the first branch completes, creating outcome-dependent companion availability.

Table 1: Recorded common settings. “Requests” excludes replayed replies and tool calls. Limits are controller-side settings, not independently verified supplier limits.

Setting	Recorded value
Pi / protocol	0.84.2 / Responses
Provider-returned model	deepseek-flash
Effort / concurrency	low / 1
Capture / suffix caps	12 / 12 newly dispatched requests
Per-phase wall limit	3,600 seconds
Context / output reservation	131,072 / 65,536 tokens
Projection eligibility	≥ 10 KiB; two full exposures; recency and error protection
Shared archive-call limit	3 successful search/read operations
Search return limit	At most 5 matches; bounded excerpts
Exact-read length	At most 6,000 Unicode codepoints; default 2,000
Budget visibility	Character bounds in tool schema; no live remaining-request or remaining-call counter in actor input

requests must match in tools, model settings, and all fields except the declared tool-result representation. Later tool choices and replies may diverge. The experimental unit is a submitted run at one realized prefix, rather than an API request or an independent sample implied by a new seed.

3.3 Order assignment and the stop rule

Within each source group, the controller sorts sibling task identifiers. A seed-and-source hash selects the initial parity, and successive siblings alternate order:

$$A_{i1} = F \iff (b_g + j_i) \bmod 2 = 0, \quad (1)$$

where b_g is the parity of SHA-256 of the seed, a zero separator, and source label g , and j_i is the sibling rank. This balances order within the original two-task source groups; it is a deterministic allocation rule, not an independently randomized assignment. Across all 76 paired/capture plans, 38 specify each initial arm.

The historical runner executes branches sequentially and *breaks when a branch’s state is not completed*. A final but incorrect answer still counts as completed for this control-flow decision. Thus a first-arm cap prevents the companion from running, whereas a first-arm wrong answer does not. Among the 27 boundary runs, ten first arms exhaust their caps and all ten companions remain unexecuted. In the other two stopped pairs, full completes first and projected then reaches its cap.

Let M_{i2} denote whether the companion is observed and C_{i1} whether the first branch completes. The scheduling mechanism has

$$\Pr(M_{i2} = 1 \mid C_{i1} = 0, A_{i1}, X_i) = 0. \quad (2)$$

The missingness is explained by an observed first-stage endpoint, but that explanation supplies no companion observations

in this stratum. Adjustment for the first endpoint alone therefore cannot recover the missing comparisons: observation positivity fails [19]. We retain the corresponding binary outcomes as unknown and bound their contribution.

3.4 Fitted selector and analysis frames

The recorded search considers 36 candidates using four completed fitting pairs and chooses the rule

$$r(X_i) = \mathbf{1}\{h_i \geq 93,641\}, \quad h_i = \text{history bytes}. \quad (3)$$

Here $r = 1$ chooses projected and $r = 0$ chooses full. The fitting tasks are `click-parameters`, `packaging-compatibility`, `boltons-indexedset`, and `tomlkit-datetimes`; all are marked in Section B and Tables 7 and 8. The threshold equals the fitting history size of `tomlkit-datetimes`.

We separate all 76 paired/capture submissions, their 27 realized boundaries, the 17 boundaries with both bounded outcomes observed, and the eleven jointly correct pairs. Six further boundary runs come from single-arm integration and are reported separately. The thirteen comparable runs outside fitting remain adaptively developed observations, not a new held-out test set.

4 Completion and Resource Estimands

4.1 Known bounded failures and missing companions

For run i and arm $a \in \{F, P\}$, define

$$Y_{ia}(K_i) = \mathbf{1}\{\text{correct final answer by request } K_i\}. \quad (4)$$

Every paired plan considered here fixes $K_i = 12$ new suffix requests. An incorrect final answer or exhaustion of these requests without an answer gives $Y_{ia} = 0$. Eventual completion with a larger allowance is a separate quantity. Unexecuted companions have unresolved Y_{ia} ; a stop before K_i would also require distinguishing a request-only endpoint from a joint token/time/request endpoint. In this 27-run frame, the unresolved companions have zero suffix requests, and the observed capped branches use all twelve.

Represent known success, known failure, and an unknown outcome by $[\ell_{ia}, u_{ia}] = [1, 1], [0, 0], [0, 1]$, respectively. For the finite-frame contrast $\Delta_N = N^{-1} \sum_i (Y_{iP} - Y_{iF})$, the usual worst-case bounds [16, 17] are

$$\Delta_N \in \left[\frac{1}{N} \sum_i (\ell_{iP} - u_{iF}), \quad \frac{1}{N} \sum_i (u_{iP} - \ell_{iF}) \right]. \quad (5)$$

Both endpoints are attainable under unrestricted unresolved binary outcomes; Section A gives the proof. These bounds describe incomplete information in a fixed recorded frame, without claiming that the frame represents deployment.

For the selector, distinguish its own outcome from harm relative to the matched full reference:

$$Y_i^r = (1 - r_i)Y_{iF} + r_iY_{iP}, \quad (6)$$

$$H_i^r = r_iY_{iF}(1 - Y_{iP}). \quad (7)$$

A selected arm capped without an answer has a known failure of Y_i^r , even when the unexecuted reference leaves H_i^r unknown. A full selection has zero selected harm relative to that same recorded full continuation by construction.

4.2 An observed jointly successful stratum

Define $J = \{i : Y_{iF} = Y_{iP} = 1\}$. Every one of the twelve incomplete pairs has at least one known zero. None can enter J under any assignment to its unexecuted companion. Consequently, J consists of exactly the eleven observed jointly correct pairs within the 27-run frame. The cost comparison on this stratum is fully ascertained; uncertainty about other arms does not change its membership.

This is a finite, recorded-coupling counterpart of principal stratification [20, 21]: cost to a correct answer is compared where both recorded continuations succeed. With adaptive task construction and a single stochastic continuation per arm, it is not an identified population survivor-average causal effect under future resampling. Lee’s selection bounds [22] require additional design assumptions and do not automatically apply. The useful result here is narrower and direct: all members of the observed jointly successful stratum are known, and its resource comparison remains distinct from whole-frame completion.

4.3 Requests, tokens, and cache categories

Let B_i be shared-prefix tokens, q_{ia} newly dispatched suffix requests, and $\bar{\tau}_{ia}$ mean reported tokens per suffix request. Then

$$S_{ia} = q_{ia}\bar{\tau}_{ia}, \quad (8)$$

$$T_{ia}^{\text{logical}} = B_i + S_{ia}, \quad (9)$$

$$T_i^{\text{acquired}} = B_i + S_{iF} + S_{iP}. \quad (10)$$

This separates shorter requests from fewer interactions and avoids treating replayed prefixes as free deployment work. Logical expenditure remains an observed partial cost when an arm ends without a correct answer; cost-to-correct-completion is reported only for J .

For $R_i = T_{iP}/T_{iF}$ on J , the ratio of sums is a full-cost-weighted mean:

$$\frac{\sum_{i \in J} T_{iP}}{\sum_{i \in J} T_{iF}} = \sum_{i \in J} w_i R_i, \quad w_i = \frac{T_{iF}}{\sum_{j \in J} T_{jF}}. \quad (11)$$

We also report the geometric mean $\exp(|J|^{-1} \sum_{i \in J} \log R_i)$ and median. The geometric mean treats reciprocal multiplicative changes symmetrically; the median describes the middle pair.

Table 2: First-arm outcomes in all 27 paired/capture boundary runs. F-first and P-first refer to the preassigned order. This complete first-arm description avoids companion missingness, but compares different adaptively obtained cases.

Outcome	F-first	P-first
Boundary runs	15	12
Correct final answer	9	5
Incorrect final answer	3	0
First-arm request cap	3	7
Cap fraction	20.0%	58.3%

Equation (11) explains the weighting of the sum ratio, not an identity equating it with the median.

Reported tokens are $U + C + O$, where U is uncached input, C cached input, and O output; C is already part of total input. With independently specified per-million prices, modeled cost is $(p_u U + p_c C + p_o O)/10^6$. The private price schedule and bill are unavailable, so we report categories and algebraic sensitivity rather than dollar savings.

5 Results

5.1 Run flow and first-arm outcomes

The submitted archive contains 86 runs and 641 completed model requests: 7,971,276 input and 231,556 output tokens, totaling 8,202,832. Of input tokens, 5,974,272 (74.9%) are reported cached. All recorded responses carry the label `deepseek-flash`; these are proxy-reported model and usage identities. The paired/capture and single-arm flows are separated in Figure 2. Forty-four paired/capture runs finish before eligibility, five stop in capture, and 27 reach a verified boundary. A long run or a prepared operation alone is insufficient evidence of a realized intervention boundary.

The 15 completed pairs are *ten F-first and five P-first*; the twelve stopped pairs are five F-first and seven P-first. Table 2 makes the first-period comparison explicit. All first arms are observed, and P-first has a higher cap fraction. Order is deterministically balanced within sources, however, and the boundary subset and source questions differ between the two order groups. These counts describe the first-arm record rather than a randomized treatment effect.

5.2 Completion before and after the filter

The fifteen completed pairs contain eleven jointly correct, two jointly incorrect, one full-only correct, and one projected-only correct result: 12/15 correct for each arm. The remaining twelve boundary runs all contain an observed cap. Retaining them produces the endpoint counts in Table 3.

The finite-frame bounds are $\Delta_{27} \in [-9/27, 1/27]$, or $[-33.3, +3.7]$ percentage points. Equivalently, *even if every unexecuted companion is resolved as favorably as possible for projection, projection can lead by at most one task out of 27*. The completed-pair tie is therefore specific to the

Table 3: Bounded outcomes over all 27 paired/capture boundaries. A cap with no correct final answer is a known zero; an unexecuted arm is unknown.

Recorded endpoint	Full	Projected
Correct	14	12
Known bounded failure	6	12
Unexecuted	7	3

filter. The interval still allows a small advantage; it is an identification interval, not a population confidence interval.

5.3 The selector’s tie depends on a fitting case

Seventeen runs have both bounded outcomes observed: the fifteen completed pairs and two full-complete/projected-capped pairs. They include all four pairs used to fit the selector. Table 4 separates those cases from the thirteen outside fitting. Every row uses exactly the same cases for the two policy choices and counts a capped selected arm as a failure. Token totals are expenditures, including failed trajectories, rather than costs to a correct answer in every case.

The selector projects five comparable cases. Two are fitting pairs: `packaging-compatibility` and `tomlkit-datetimes`. The other three are `filelock-contention`, `more-itertools-window`, and `pathspect-util`. On the fitting set, the packaging success removes one full-arm failure. Outside fitting, the selected `pathspect cap` adds one. Thus equality at 3/17 is not evidence of equal performance outside fitting.

A hindsight threshold-class optimum, minimizing observed failures and then expenditure, chooses 95,519 bytes—exactly the fitting history size of `packaging-compatibility`. It yields two failures on the same seventeen cases and 2.8% less suffix expenditure (2.0% less logical expenditure) than always-full. This is an in-sample reference within this threshold family, not a general upper limit on possible agent savings. In leave-one-source-out development analysis, twelve of fourteen folds choose 95,519, and the concatenated held-out choices incur four failures versus three for the frozen rule. Full policy comparisons appear in Table 11; Section C gives the strict-format sensitivity.

5.4 Joint success: less aggregate input, more interaction

The observed $J = 11$ success stratum is fully determined within the 27-run frame. Its logical totals are 1,267,036 tokens for full and 949,774 for projected: a ratio of 0.750, or 25.0% less in aggregate. The geometric mean ratio is 0.883, whereas the median is 1.292; seven of eleven pairs increase. Figure 3 shows every pair and marks fitting cases. The largest absolute saving occurs in `more-itertools-recipes`; removing that one case in a leave-one-out sensitivity changes the sum ratio to 1.077. The case remains in the primary analysis.

Request counts explain why a shorter representation may

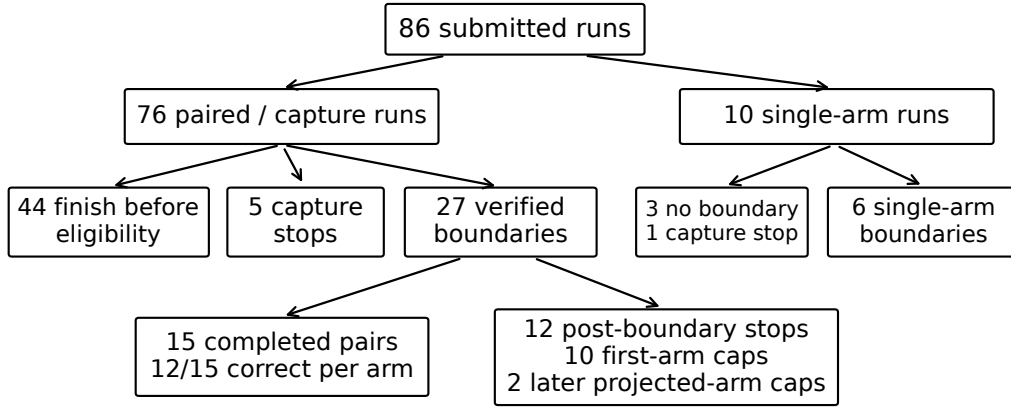


Figure 2: Accounting for all 86 submitted runs. The paired/capture frame contains 27 boundaries: 15 completed pairs and 12 stopped pairs. Ten stops occur in the first arm, suppressing the companion; two occur after full has completed. The separate single-arm family contributes six additional boundaries.

Table 4: Frozen selector versus always-full, stratified by fitting participation. Tokens include each run’s shared prefix once. The overall failure tie is sustained by the fitting case `packaging-compatibility`, where projected is correct and full is wrong. “Outside fitting” denotes already observed development runs, not an untouched test set.

Frame	Runs	Full failures	Selector failures	Full logical tokens	Selector logical tokens	Selector – full
Fitting pairs	4	1	0	404,703	298,261	–106,442 (–26.3%)
Outside fitting	13	2	3	1,419,197	1,541,879	+122,682 (+8.6%)
All comparable	17	3	3	1,823,900	1,840,140	+16,240 (+0.9%)

Table 5: Reported token categories for $J = 11$, including the shared prefix once per logical arm. Cached input is a subset of input, not an additional token charge. Full per-pair categories are in Table 10.

Category	Full	Projected
Uncached input	264,132	353,816
Cached input	982,272	574,592
Output	20,632	21,366
Total	1,267,036	949,774

take more work. Projected uses more suffix requests in eight pairs, fewer in two, and the same number in one. Totals are *55 versus 35 requests*, a 57% increase. This is the interaction dimension of Equation (8), separate from the token dimension. The two-sided sign-test calculation for seven token increases out of eleven is $p = 0.55$; it is a descriptive reference under independent signs, not a claim of general harm from these related development cases.

Projection is accompanied by 89,684 more uncached input tokens, 407,680 fewer cached tokens, and 734 more output tokens on J (Table 5). Rewriting an earlier request prefix can reduce cache reuse, while changed continuations add further differences. These records show the category changes, not an isolated causal effect on the private cache. The modeled monetary difference is $(89,684p_u - 407,680p_c + 734p_o)/10^6$; its sign depends on prices. Earlier, separately recorded one-prefix

evidence shows the same qualitative issue with different counts, and is kept out of this campaign’s aggregates (Section F).

5.5 A retrieval path that still exhausts the budget

For `pathspect-util`, history size is 94,493 bytes, so the frozen selector chooses projected. Full completes correctly in three suffix requests and 84,202 suffix tokens. Projected consumes twelve requests and 237,629 suffix tokens without a final answer. Mean tokens per suffix request are therefore approximately 28.1k for full and 19.8k for projected: 29% less per request but four times as many requests. Its total suffix expenditure is 2.82 times full. Including the 39,539-token common prefix yields 123,741 full tokens and 277,168 projected tokens spent.

After nine common-prefix reads, full issues five further reads. Projected issues thirteen reads, two searches, and three exact-read requests. The first exact read and both searches return archived content; the next two exact reads receive the shared-call-quota refusal. These refusal payloads are recorded with `isError=false`. This reflects an extension-runtime mismatch: the recovery tool returned `isError: true` in its result object, whereas the pinned Pi loop marks a tool failure only when `execute` throws [26]. The recorded model-visible text remained `ReVer halted: recovery_quota:` its error flag is absent from the Responses tool-output items,

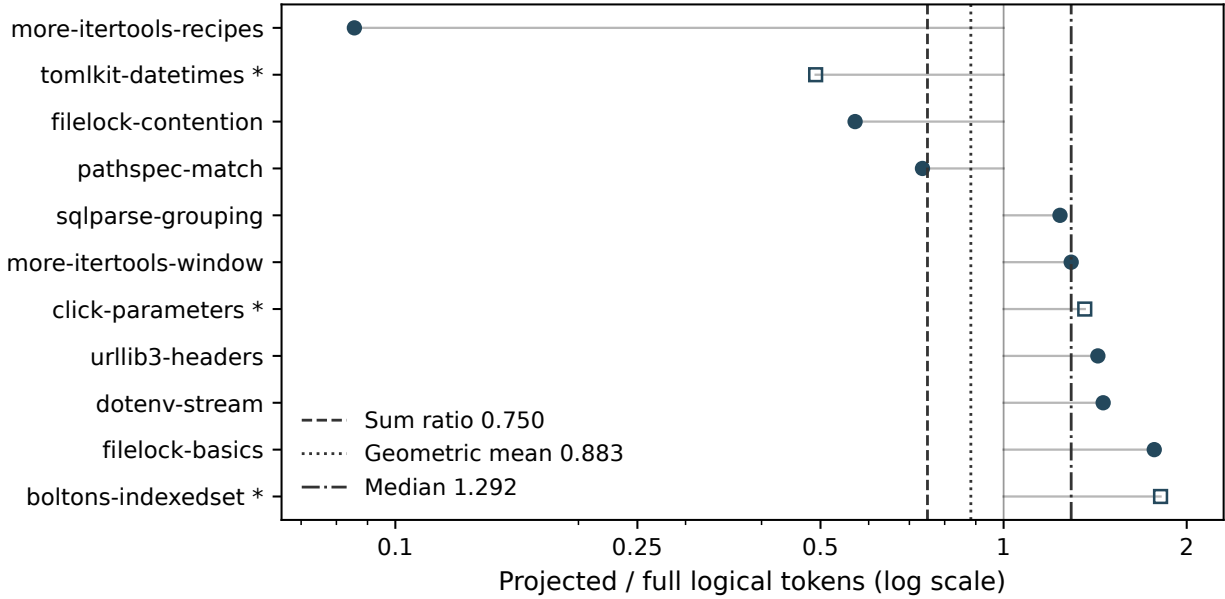


Figure 3: Logical token ratios for the fully observed jointly correct stratum. All data marks share one color; hollow squares and asterisks identify fitting pairs. The sum ratio, geometric mean, and median summarize different aspects of this heterogeneous distribution.

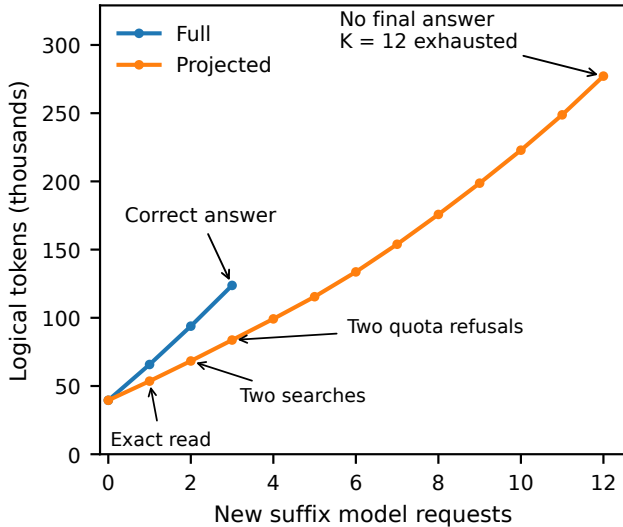


Figure 4: Recorded pathspec-util expenditure. The projected branch successfully reads archived text after request 1 and searches after request 2. Two subsequent exact-read requests receive quota refusals after request 3. Model requests continue to the cap; the last point is expenditure without a final answer.

and these short archive-tool results are protected from projection. The mismatch was fixed after the campaign in pull request 15 [27]; the recorded trajectories remain unchanged. The actor then continues reading source fragments and reaches the suffix cap (Figure 4). The evidence establishes retrieval

success and subsequent quota refusal, but neither the sufficiency of the retrieved fragments nor the outcome of a hypothetical larger quota.

5.6 Trajectory features vary across repeated tasks

Six task identities appear at boundaries in both paired and later single-arm runs (Table 9). `more-itertools-recipes` changes from 56,382 to 110,194 history bytes; `pathspec-match` changes from 63,576 to 97,045. Both decisions flip from full to projected. The latter is the only capped single-arm continuation and shares the pathspec source group with the paired adverse case. These are six descriptive repeated trajectories: history size is a property of the captured interaction, not a fixed task attribute. The single-arm runs provide no missing alternative-arm result.

6 Discussion and Scope

Complete both allocated continuations. The first design lesson concerns the runner, before the selector. Once a valid boundary has been captured, each allocated arm should have a separate, preauthorized resource reservation and an independent execution decision. A first-arm wrong answer or ordinary request-cap exhaustion should not suppress its companion. Integrity faults, revoked permission, and unsafe environments remain valid reasons to stop; their unresolved outcomes must be visible. Balanced initial order alone cannot repair a completion-dependent branch suppression rule.

Table 6: Claims and scope of this methodological case study.

Object	Supported result and scope limit
Run process	Known order, boundaries, caps, and missing companions; not randomized population coverage.
Joint success	Eleven fully identified recorded pairs and their costs; not a population always-success effect.
Selector	Fitting and outside-fitting behavior of one frozen rule; not generalization or deployment permission.
Resources	Reported requests, token categories, and archive returns; not dollar savings or retrieval necessity.

Treat opportunities and outcomes separately. Forty-four of 76 paired/capture runs finish without a boundary. The qualification rule requires sufficiently large, repeatedly exposed observations, so it favors trajectories that have already incurred repeated input. This creates a tension: the cases offering more repeated text to remove may also require more remaining reasoning and retrieval. The current development frame exhibits both rare opportunities and capped eligible continuations; it does not estimate their association in a deployment population. Forcing a task to make extra reads just to create eligibility would change the efficiency question.

Make resource signals usable. The quota example distinguishes storage availability, valid tool access, and an answer within budget. Here, a returned error property was ignored by the pinned Pi loop; throwing now aligns the structured status with the refusal. The historical model still saw the refusal text, so correcting the flag alone supplies no evidence of improved task completion. A successful search may already supply enough text, making mandatory exact reading unnecessary. Allowance visibility and quota allocation remain distinct interventions; changes should be tested symmetrically rather than used to repair the recorded cap retrospectively.

Separate fitting, stress cases, and confirmation. The selector’s extra-fitting failure and its threshold’s dependence on fitting values argue against promoting the aggregate tie to a generalization result. Existing cases remain useful development evidence, including the genuine large savings. Moving the cutpoint above the adverse history size would be another fit on known observations. A future quality-constrained objective should retain known bounded failures and unresolved policy outcomes before comparing resource use.

Limits and reproducibility. Source questions were built adaptively on installed package subsets, often with two questions per source and repeated histories. The proxy-reported model label does not authenticate backend weights. Each arm provides one stochastic continuation, the read-only adapter is not the complete Pi product, and source labels do not establish independence. Historical extracted-answer scoring remains the primary contract; a strict whole-object sensitivity adds

one failure to each arm without changing the selector comparison (Section C). The article’s ancillary files contain analysis inputs, generation scripts, extracted examples, and artifact digests (Section E). Raw private requests, session databases, provider configuration, and invoices are not distributed. Integrity checks establish consistency rather than experimental chronology. Table 6 distinguishes the finite recorded findings from deployment claims.

7 Conclusion

Completed pairs alone show 12/15 success for each arm, yet the underlying stop rule removes ten companions after a first-arm cap. Including every realized boundary leaves a range of -9 to $+1$ projected-minus-full successes across 27 runs. The eleven jointly correct pairs support a precise but stratum-specific cost result: fewer aggregate tokens coexist with more requests and a higher median token ratio. The frozen selector’s overall failure tie includes its fitting data; on thirteen comparable runs outside fitting, it fails once more and spends 8.6% more logical tokens. ReVerPi thus provides a concrete methodological case: evaluate bounded completion on the complete intervention frame, preserve missing companions, report interaction as well as token use, and keep fitting performance separate from evaluation.

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A Formal Details and Interpretation

A.1 Finite-frame bounds and the observed success stratum

For each run i , only the unresolved binary coordinates of (Y_{iF}, Y_{iP}) vary. The smallest contribution to $Y_{iP} - Y_{iF}$ is $\ell_{iP} - u_{iF}$ and the largest is $u_{iP} - \ell_{iF}$. Under no cross-run restrictions, these extremes can be assigned simultaneously. Summing proves Equation (5) and its sharpness. This is the standard worst-case construction of partial identification [16, 17]; independence is unnecessary for this finite-frame calculation.

The interval does not mean that every particular intermediate real-valued point is attainable: with $N = 27$, the feasible means are on the integer-over-27 lattice. Its endpoints are attainable. Full has fourteen known successes and seven unresolved companions, so its possible success count is 14–21; projected has twelve known successes and three unresolved companions, so its count is 12–15. The contrast therefore ranges from $12 - 21 = -9$ to $15 - 14 = 1$ successes.

Membership in J has a simpler identification argument. The fifteen completed pairs identify eleven $(1, 1)$ outcomes and four other outcomes. Each of the twelve stopped pairs has an observed zero. Thus all 16 nonmembers of J are ruled out, including the ten with unexecuted companions. No missing outcome can change the membership of the observed jointly successful stratum. This justifies comparing successful-task costs within these eleven recorded pairs. It does not identify a latent population stratum under repeated stochastic potential continuations or a randomized treatment assignment. The connection to principal stratification [20, 21] is a useful estimand distinction, not a claim that the campaign estimates a population SACE. Lee-style trimming [22] would require assumptions about assignment and monotone selection that are not warranted by this record.

A.2 Selected outcomes and source grouping

For a fixed binary rule r_i , the sharp bounds on its paired selected harm are

$$\ell_i^H = r_i \ell_{iF}(1 - u_{iP}), \quad u_i^H = r_i u_{iF}(1 - \ell_{iP}). \quad (12)$$

A projected success or full failure rules out harm. A full selection also sets harm to zero relative to the recorded full reference, even if the projected arm was never run. For a fixed source group g with a declared set of tasks I_g , an any-harm event is $H_g = \max_{i \in I_g} H_i^r$. Its bounds are $[\max_i \ell_i^H, \max_i u_i^H]$, and their means over the declared groups bound the finite-frame any-harm fraction. The event depends on group size and task composition; a source label alone does not establish comparable independent draws.

A post hoc nonfitting frame of 22 paired/capture boundary runs contains seventeen source labels, one known harmful group, and three unresolved groups. Its any-harm fraction is bounded by $[1/17, 4/17]$, approximately $[5.9\%, 23.5\%]$. This source-level identification interval answers a different question from the 27-task projected-minus-full contrast.

A.3 Why observation adjustment cannot supply absent companions

Let D_i indicate inclusion in a completed-pair table. For any fully specified finite vector of harms H_i , the difference between the retained and complete-frame means is

$$\overline{H}_{D=1} - \overline{H} = \frac{\text{Cov}_N(H, D)}{\overline{D}}, \quad \overline{D} > 0. \quad (13)$$

This follows by expanding $\text{Cov}_N(H, D) = N^{-1} \sum_i H_i D_i - \overline{H} \overline{D}$. The covariance need not have a universal sign. In the present case, the observed selected cap has $H_i = 1$ and $D_i = 0$, demonstrating a relevant exclusion.

The observation process is known: after a first-arm noncompletion, the controller breaks. Treating this as explained by the observed first-arm result does not restore overlap. In the capped-first-arm region, the probability of observing the companion is exactly zero under the implemented schedule. Inverse-observation weighting would divide by zero there, and a fitted regression would rely entirely on extrapolation into an unobserved region. The worst-case interval leaves that extrapolation unspecified [18, 19].

For a hypothetical target distribution, overall selected harm and harm conditional on projection additionally satisfy

$$\Pr(H^r = 1) = \Pr(r = 1) \Pr(Y_F = 1, Y_P = 0 \mid r = 1). \quad (14)$$

Low selection can make overall harm small while supplying little evidence about projected cases, as in selective prediction [28]. The campaign's adaptive frame supplies descriptive counts, not estimates of these deployment probabilities.

B Complete Recorded Tables

Table 7: All fifteen completed pairs. An asterisk identifies one of the four fitting tasks. T is logical prefix-plus-suffix expenditure, q suffix requests, and Y the historical extracted-answer score. Costs in rows with an incorrect answer are expenditure, not cost to a correct answer.

Task	First	History bytes	Y_F/Y_P	q_F/q_P	T_F	T_P	T_P/T_F
click-parameters*	F	62,102	1/1	3/8	76,016	103,353	1.360
packaging-compatibility*	F	95,519	0/1	1/1	57,171	51,359	0.898
boltons-indexedset*	F	52,266	1/1	2/6	73,859	133,797	1.812
cerberus-errors	F	52,968	0/0	2/5	48,494	63,500	1.309
dotenv-stream	P	25,110	1/1	2/5	26,662	38,865	1.458
filelock-basics	F	72,397	1/1	3/11	91,143	161,204	1.769
filelock-contention	P	96,013	1/1	3/4	114,864	65,465	0.570
funcky-colls	F	49,128	0/0	2/4	44,800	52,456	1.171
more-itertools-recipes	P	56,382	1/1	9/1	418,364	35,826	0.086
more-itertools-window	F	110,836	1/1	1/2	63,993	82,647	1.292
pathspec-match	P	63,576	1/1	2/2	66,448	48,875	0.736
six-moves	F	39,668	1/0	2/5	51,560	79,484	1.542
sqlparse-grouping	P	89,881	1/1	2/6	75,974	94,050	1.238
tomlkit-datetimes*	F	93,641	1/1	6/5	197,657	97,027	0.491
urllib3-headers	F	62,556	1/1	2/5	62,056	88,665	1.429

Table 8: All twelve stopped paired/capture boundary runs. A question mark denotes an unexecuted arm; an observed capped arm has $Y = 0$. Rule selects projected when history bytes are at least 93,641. In ten rows the first arm caps and the companion is unexecuted; in the other two, F completes and P subsequently caps.

Task	First	History bytes	Rule	Y_F/Y_P	q_F/q_P
attrs-define	P	58,749	F	?/0	0/12
docutils-nodes	F	72,019	F	1/0	8/12
docutils-tree	P	154,876	P	?/0	0/12
markdown-extensions	P	130,201	P	?/0	0/12
marshmallow-fields	P	51,348	F	?/0	0/12
pathspec-util	F	94,493	P	1/0	3/12
pyparsing-helpers	P	81,100	F	?/0	0/12
pyparsing-parse	F	45,227	F	0/?	12/0
rich-table	F	62,360	F	0/?	12/0
sqlalchemy-core	F	48,268	F	0/?	12/0
tabulate-formats	P	124,625	P	?/0	0/12
tomlkit-roundtrip	P	64,489	F	?/0	0/12

Table 9: All six single-arm boundary runs and the earlier paired boundary for the same task. These are repeat trajectories, not paired counterfactuals for the single-arm run. Two decisions change from F to P. The single-arm pathspec-match ends without an answer; its expenditure is partial.

Task	Paired bytes	Single-arm bytes	Decision	q	Endpoint	Logical tokens
filelock-basics	72,397	77,200	F $\rightarrow$ F	2	correct	79,050
filelock-contention	96,013	123,601	P $\rightarrow$ P	10	correct	206,300
more-itertools-recipes	56,382	110,194	F $\rightarrow$ P	2	correct	100,078
pathspec-match	63,576	97,045	F $\rightarrow$ P	12	cap exhausted	244,370
sqlparse-grouping	89,881	64,817	F $\rightarrow$ F	5	correct	135,759
urllib3-headers	62,556	54,818	F $\rightarrow$ F	2	correct	49,185

Table 10: Complete token categories on the observed jointly correct stratum J . U is uncached input, C cached input, O output, and q suffix requests. Categories include capture once for each logical arm. Fitting pairs are marked with an asterisk.

Task	U_F	C_F	O_F	U_P	C_P	O_P	q_F	q_P
more-itertools-recipes	60,479	352,768	5,117	15,514	18,944	1,368	9	1
tomlkit-datetimes*	28,063	167,424	2,170	42,956	52,480	1,591	6	5
filelock-contention	24,917	87,552	2,395	24,380	38,528	2,557	3	4
pathspec-match	22,217	41,472	2,759	28,724	17,792	2,359	2	2
sqlparse-grouping	24,755	49,152	2,067	46,517	45,056	2,477	2	6
more-itertools-window	29,340	33,792	861	44,430	36,992	1,225	1	2
click-parameters*	18,965	55,424	1,627	37,691	62,848	2,814	3	8
urllib3-headers	14,984	45,952	1,120	27,242	60,032	1,391	2	5
dotenv-stream	5,434	20,736	492	9,660	28,032	1,173	2	5
filelock-basics	21,755	68,736	652	56,096	102,656	2,452	3	11
boltons-indexedset*	13,223	59,264	1,372	20,606	111,232	1,959	2	6

C Selector and Scoring Sensitivity

Table 11: Policy comparisons on the same seventeen runs with both bounded outcomes known. The common prefix totals 510,392 tokens. The hindsight oracle is the observed lexicographic optimum within the history-threshold family, minimizing failures and then tokens; it is neither a deployed policy nor an upper bound on all possible agent improvements.

Policy	Failures	Suffix tokens	Logical tokens	Logical change vs F
Always-full	3	1,313,508	1,823,900	—
Always-projected	5	1,240,753	1,751,145	−4.0%
Frozen threshold 93,641	3	1,329,748	1,840,140	+0.9%
Hindsight threshold 95,519	2	1,276,951	1,787,343	−2.0%

The frozen threshold equals the history of `tomlkit-datetimes`, a fitting pair. The hindsight value 95,519 equals `packaging-compatibility`, another fitting pair, and lies above the adverse `pathspec-util` history of 94,493. In fourteen leave-one-source-out development folds, the fitted thresholds are 95,519 in twelve folds, 96,013 in one, and 93,641 in one. The held-out predictions have four failures, compared with three for the frozen rule. Each fold holds out a whole source group, so it trains on a variable number of the remaining runs rather than always sixteen. All folds still reuse this observed development frame.

Scoring sensitivity. The historical contract extracts the first decodable JSON object from a final answer. Keeping the same seventeen-run denominator, a stricter whole-object contract gives four full failures, six projected failures, and four frozen-selector failures, compared with three, five, and three under the historical contract. The ordering is unchanged. This alternative accepts a whole JSON object, optionally enclosed in a whole JSON fence, and rejects prose wrappers, duplicate keys, and nonfinite values before normalization. Three historical final-answer labels differ across the entire campaign: both `click-parameters` arms and the no-boundary `packaging-normalization` capture.¹ Historical main-table scores remain unchanged.

Aggregate and multiplicative summaries. On J , the geometric mean is 0.883, the median 1.292, the sum ratio 0.750, and the arithmetic mean of ratios 1.113. Exchanging F and P reciprocates the geometric mean and sum ratio, while the arithmetic mean of reciprocal ratios is generally not the reciprocal of the original mean. The median and sum ratio can disagree because they target the middle observation and a cost-weighted mean, respectively; Equation (11) is specifically a weighted-mean identity. Seven of eleven token ratios exceed one. A conventional two-sided binomial sign calculation gives $p = 0.55$, which is only a reference calculation because tasks share source groups and the frame was adaptively assembled.

¹An implementation-level counterexample is `{"a":0,"a":1}`: a permissive decoder can discard the first key before a strict scorer sees it. This counterexample does not establish that any observed successful answer contained duplicate keys.

D Concrete Task and Projection Example

The following is the original program in the `pathspec-util` question, using the recorded `pathspec` 1.1.1 source snapshot. The task asks for the six named fields and exactly one JSON object. The actor receives the program and allowlisted source, not the controller's answer. The listing preserves the original question verbatim: its final comment names `pathspec.__version__` without a separate `import pathspec`. That comment specifies a requested field from the supplied source snapshot; it is not an executable expression in this snippet. The historical question and scoring are unchanged.

```
from pathspec import PathSpec
from pathspec.util import normalize_file
other = PathSpec.from_lines("gitwildmatch", ["docs/*", "/root.txt"])
a = other.match_file("docs/index.md")
b = other.match_file("docs/sub/page.md")
c = other.match_file("root.txt")
d = other.match_file("sub/root.txt")
e = normalize_file("dir\\file.py")
# Required fields: docs_index=a, docs_nested=b, root_txt=c,
# sub_root_txt=d, normalize=e, version_major=pathspec.__version__.split(".")[0]
```

The controller reference, reproduced here solely to specify the scoring example, is:

```
{"docs_index":true,"docs_nested":true,"normalize":"dir\\file.py",
 "root_txt":true,"sub_root_txt":false,"version_major":"1"}
```

The first projected request actually contains the following kind of replacement for a 25,551-byte tool observation. The display below abbreviates the hash and head/tail blocks for typesetting; bracketed editorial ellipses are not additional text sent to the model. The complete recorded replacement is included as `anc/data/actual_observation.txt` in the accompanying ancillary files.

```
[reverpi-observation-v1 {"handle":"6c69f25c...5833d22ea",
 "offset_unit":"unicode_codepoints","original_bytes":25551,
 "tool":"read","total_chars":25551}]
Historical output omitted from this request, not deleted. This is not a current-state verification.
Read exact historical text with recover_evidence(handle, start, chars).
Locate a span with search_evidence(query, chars); use returned handle and start.
Never send query to recover_evidence.
[head]
"""
This module provides utility methods for dealing with path-specs.
"""
from __future__ import annotations
[editorial elision of remaining head excerpt]
[middle omitted]
[editorial elision of leading tail excerpt]
def stat(self, follow_links: Optional[bool] = None) -> os.stat_result:
[editorial elision]
return self._stat if follow_links else self._lstat
[tail]
```

The excerpt budget is a total head/tail allowance rather than the size of the entire replacement, which also contains metadata and instructions. The handle resolves to the original observation, not to the truncated display. Search returns at most five matches, each with a bounded original-text excerpt. The successful search/read records count against a combined limit of three. After that limit, the two refused exact reads are represented in the historical transcript as error text with a false error flag. The plot annotates those tool events at the model response that requested them, rather than treating tool calls as extra model requests.

E Source Inventory, Record Integrity, and Availability

The campaign archive is identified by SHA-256

2b57e3188492a6b852684e8a85a19b92db96e7442fbb825a891ae762cf22f29c.

Its 12,980 listed artifacts support the original request-level census. Raw response counts and usage are cross-checked with SQLite attempts, and phase endpoints with plan limits and sealed boundaries. The source-level auditor reports 73 passes among the 76 paired/capture runs; three retain their allowlist-intent rejection. Of the passes, 71 match a submitted audit and two have no submitted counterpart. Accounting completeness and interface compliance are distinct checks; the rejected interface cases are retained in the census.

Table 12: Fixed source subsets for the 76 paired/capture submissions. Each source contributes two investigator-authored task plans, with balanced initial order within that source. “Boundary” and “Pair” count realized boundaries and completed pairs, respectively. File counts include the subset’s license files; shortened snapshot digests are locators, with complete digests in the supplementary data.

Source/version	Files	Boundary	Pair	Snapshot SHA prefix
attrs-26.1.0	20	1	0	fc4d96802c3d
boltons-26.2.0	31	1	1	b80f6739e270
cachetools-7.1.4	9	0	0	1fa516c2685d
cerberus-1.3.8	7	1	1	f4bb235519e6
click-8.1.8	17	1	1	44353e20a2fd
colorama-0.4.6	14	0	0	be58df74478d
dateutil-2.9.0.post0	19	0	0	a04fe4c3b097
docutils-0.23	130	2	0	95cdc9b6127e
filelock-4.0.3	24	2	2	6cb9581b6f10
funcy-2.1	17	1	1	f0aa75f84a32
humanize-4.16.0	8	0	0	8da0a5d02e67
idna-3.20	11	0	0	1a074bdb5b8c
jinja2-3.1.6	26	0	0	86c4faf0e95b
jmespath-1.1.0	9	0	0	c21828454ba7
markdown-3.10.3	34	1	0	befd520129ae
marshmallow-4.3.1	15	1	0	7b6f357d354c
more-itertools-11.1.0	4	2	2	c5fa441afd8a
networkx-3.7	584	0	0	c4992489e474
packaging-25.0	19	1	1	c8879839459f
pathspec-1.1.1	32	2	1	db9ce114b658
pip-26.2.1	405	0	0	07eb31e79304
platformdirs-4.11.2	10	0	0	7cc2d6b8dc56
pygments-2.21.0	344	0	0	6fe831a0b547
pyparsing-3.3.3	18	2	0	46640a34b12e
python-dotenv-1.2.3	9	1	1	d2deeda0f0ed
rich-15.0.0	101	1	0	acdbd0ecaa80
six-1.17.0	2	1	1	713806dd6930
sortedcontainers-2.4.0	5	0	0	150744eb5936
sqlalchemy-2.0.54	258	1	0	6e6789fc1543
sqlparse-0.6.0	22	1	1	71bfdfa9998d
tabulate-0.10.0	2	1	0	4f0bccf42996
toml-0.10.2	6	0	0	102b96e7a074
tomlkit-0.15.1	13	2	1	193454b37de5
toolz-1.1.0	32	0	0	ec0bd7c7eba9
urllib3-2.8.0	37	1	1	d7d96f512c08
voluptuous-0.16.0	7	0	0	d8c89cb9ac84
wheel-0.48.0	16	0	0	c6693b2738e6
xmltodict-1.0.4	2	0	0	2e0f2b772f66

This paper’s main results preserve the declared historical extracted-answer contract. The accompanying ancillary data include branch order, history size, fitting membership, request counts, cache categories, source digests, and the strict-format sensitivity. Project code is located at https://github.com/timwhitez/ReVer_Pi. The implementation identities are pinned below; each full hash links to its repository object.

Historical reference	7a19d72ff86655310a4b24298191ce73cbd2c44e
Error-signaling fix	59e8497a56faddff8c0d1836456b3cd3cf5a4b6a
Pull request 15 merge	d331a4801873c3957b9ca410de43c520582197e4

These later code changes do not alter the archived experimental requests or results.

Data availability. The article’s source archive contains `anc/README.md`, analysis scripts, the aggregated inputs used for the tables and plots, the original task/example fields, and `anc/data/actual_observation.txt`. These materials reproduce the paper’s reported calculations and figures without new provider calls and are supplied with this article rather than promised at a future URL. The complete raw request/response archive, private provider settings, session and ledger databases, and invoices are

not distributed. The ancillary files do not provide a substitute for independently replaying the private service. Their digests establish file consistency, not experimental chronology, backend weights, or billing. Configuration variants include different token-admission budgets and task revisions; all paired suffix limits in this frame remain twelve requests.

F Calibration Context and Separate Earlier Evidence

The historical calibration keeps eleven completed rows in nine source groups, with zero selected harms. Only two rows actually select projection. Under independent, fixed-sample Bernoulli assumptions, the one-sided 95% upper bound for zero events is $1 - 0.05^{1/9} \simeq 0.283$ [25]. Risk control requires a suitable frozen procedure and calibration design [23]. This adaptive, complete-case sample does not acquire a population guarantee from reproducing the formula.

Figure 5 contrasts the following reference calculations. For illustration, a conservative sequence allocates $\alpha_n = \alpha/[n(n+1)]$ to the n th fixed-sample upper bound. Since $\sum_{n \geq 1} \alpha_n = \alpha$, a union bound gives simultaneous coverage under the relevant independent Bernoulli assumptions. With zero events at $n = 9$, the upper value is approximately 0.565 rather than 0.283. This is a simple error-spending construction, not the efficient confidence-sequence method of Howard et al. [24]; it cannot cure absent companions or adaptive question selection.

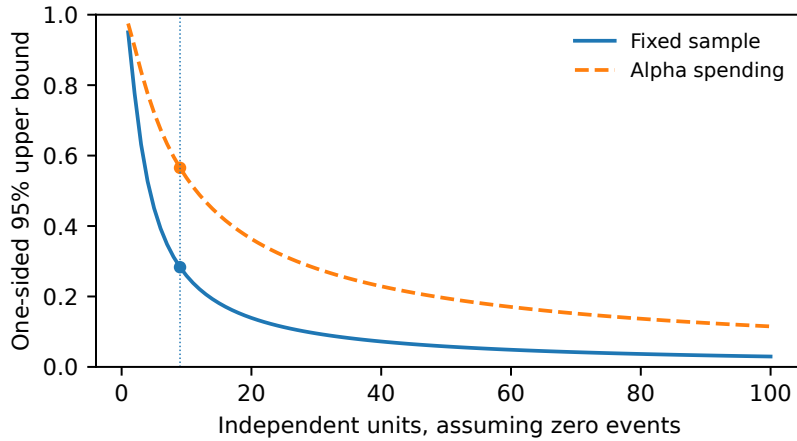


Figure 5: Illustrative zero-event upper bounds under stated sampling assumptions, not a calibration result for this campaign. Each $n = 9$ marker uses the color of its own curve.

An earlier, separate single-prefix experiment used a proxy returning a different model label. Its two correct logical trajectories used 113,462 and 86,162 tokens, with three and four suffix requests. The projected-minus-full category changes were +38,727 uncached input, -66,048 cached input, and +21 output tokens. These observations illustrate the same accounting issue as Table 5, but are not pooled into the 86-run campaign, its stratum totals, or its quality analysis.